testbook PASS PRO MAX

Quantitative Aptitude

Formula – Guide

Permutation & Combination

Permutation and Combination

Factorial Notation

- $n! = n(n-1)(n-2) \ldots\ldots\ldots\ldots 3.2.1$
- $n! = n.(n-1)!$
- $0! = 1! = 1$
- $(2n)! = 2n\,.n![1.3.5.7 \ldots\ldots\ldots (2n-1)]$
- $n! = n(n-1)! = n(n-1)(n-2)! = n(n-1)(n-2)(n-3)!$
- $n(n-1) \ldots\ldots\ldots (n-r+1) = \frac{n!}{(n-r)!}$

Permutation

- $P(n, r) = {}^{n}P_{r} = \frac{n!}{(n-r)!} = n(n-1)(n-2) \ldots\ldots\ldots (n-r+1)$
- The no. of permutations of n different objects taken r at a time when each object may be repeated any number of times is $\mathbf{n^r}$.

Permutation of alike objects:

The number of permutations of n objects taken all at a time in which, p are alike objects of one kind, q are alike objects of second kind & r are alike objects of a third kind and the rest (n – (p + q + r)) are all different is $\frac{n!}{p!q!r!}$.

Permutation under restrictions:

- The number of permutations of n different objects, taken r at a time, when a particular object is to be always included in each arrangement, is $r.\,{}^{n-1}P_{r-1}$
- The number of permutations of n different objects, taken r at a time, when a particular object is never taken in each arrangement is ${}^{n-r}P_{r}$.
- String method: The number of permutations of n different objects, taken all at a time, when m specified objects always come together is $m! \times (n-m+1)!$.

Combination

- ${}^{n}C_r = \frac{n!}{r!(n-r)!}$, $(0 \le r \le n) = {}^{n}P_r / r!$
- ${}^{n}C_r = {}^{n}C_{n-r}$
- ${}^{n}C_r + {}^{n}C_{r-1} = {}^{n+1}C_r$
- ${}^{n}C_x = {}^{n}C_y \Rightarrow x = y$ or $x + y = n$
- If n is even, then the greatest value of ${}^{n}C_r$ is ${}^{n}C_{n/2}$.
- If n is odd, then the greatest value of ${}^{n}C_r$ is $C\left(n, \frac{n+1}{2}\right)$
- ${}^{n}C_0 + {}^{n}C_r + \ldots\ldots\ldots\ldots + {}^{n}C_n = 2^n$
- ${}^{n}C_n + {}^{n+1}C_n + {}^{n+2}C_n + \ldots\ldots\ldots + {}^{2n-1}C_n = {}^{2n}C_{n+1}$

Combinations of Alike Objects:

- The number of combinations of n identical objects taking $(r \le n)$ at a time is 1.
- The number of ways of selecting r objects out of n identical objects is n + 1.
- If out of (p + q + r + s) objects, p are alike of one kind, q are alike of a second kind, r are alike of the third kind and s are different, then the total number of combinations is : $(p + 1)(q + 1)(r + 1)2^s - 1$

Combinations under Restrictions:

- Number of ways of choosing r objects out of n different objects if p particular objects must be excluded = ${}^{n-p}C_r$
- Number of ways of choosing r objects out of n different objects if p particular objects must be included $(p \le r)$ = ${}^{n-p}C_{r-p}$
- The total number of combinations of n different objects taken one or more at a time = $2^n - 1$

Circular Permutation:

- The number of circular arrangements of n different objects

$$= \frac{\text{the number of linear arrangements of } n \text{ different objects}}{n} = \frac{n!}{n} = (n-1)!$$

- **Clockwise and Anticlockwise Arrangements –**

  When clockwise and anticlockwise arrangements are not different, the number of circular arrangements of n different objects is (n – 1)!/2

- The number of circular permutations of n different objects taken r at a time

  ^nP_r /r , when clockwise and anticlockwise orders are treated as different.

  ^nP_r /2r, when clockwise and anticlockwise orders are treated the same.

- The number of circular permutations of n different objects altogether

  ^nP_n /n = (n – 1)! , when clockwise and anticlockwise order are treated as different,

  ^nP_n /2n = 1/2(n – 1)! , when the above two orders are treated as the same.